

Transport-First Defect Telemetry on Replay Artifact Graphs

A Minimal Gauge-Covariant Formulation of Closure Inconsistency

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Abstract

We study a transport-first formulation for reading locally plausible but globally inconsistent outputs on replay artifact graphs. Rather than treating reliability as a scalar score defined directly on outputs, we model each observed node as a projector-first local object, define edge-wise transport between neighboring local objects, and read closure inconsistency on explicit cycles. The cycle transport is gauge-covariant under local basis changes, while the Frobenius closure defect is gauge-invariant. For clarity, the main exposition is given in the equal-rank nonsingular case; unequal-rank overlap is handled in Appendix A through compact-SVD-supported partial isometries. This note fixes a minimal mathematical core of the protocol and states its present limits without claiming universal inconsistency detection, graph-wide operators, or field-level aggregation.

1 Problem and Positioning

Many practical evaluation pipelines for language-model outputs still rely on scalar-first readouts: confidence, likelihood, similarity, agreement, or single-score judgments. These quantities can be useful, but they are often pointwise or aggregate readouts rather than structural closure observables. They do not, by themselves, specify whether a structured inference path remains globally transport-consistent when the output is read as a relation-bearing object rather than as an isolated string.

This limitation becomes especially visible in replay-heavy and retrieval-heavy settings. A generated answer may remain locally plausible at each step, while the induced relations among support, conflict, and answer artifacts fail to assemble into a globally consistent structure. In such cases, the principal failure is not necessarily a low-confidence token or a visibly anomalous span. The relevant structural signal is read here as a *closure residual*: a declared cycle of local relations does not return consistently under induced transport.

This note proposes a transport-first alternative. Rather than asking for a better scalar confidence surrogate, we define a minimal observation protocol in which:

1. local artifact states are represented as projector-based objects,

2. directed transport is induced between these local objects,
3. explicit cycles are declared on a replay-artifact graph, and
4. inconsistency is read out as nontrivial closure defect on those cycles.

The intended contribution is therefore narrower than a general inconsistency-detection framework. We do not claim to solve inconsistency detection in the large. We claim instead that, under a declared replay protocol, one can construct a gauge-covariant, transport-first telemetry layer that identifies where a declared cycle fails to close. Declared reference structures, when present, serve only to organize residual-cycle analysis. The companion empirical report records the protocol-specific evidence base to which this formal note is scoped [1].

The goal is not to replace one scalar score by another, but to change the observational primitive. The primary public object is not a score. It is a local subspace object together with induced transport and explicit cycle structure on a replay-artifact graph. Scores, when reported, are secondary readouts of closure behavior on top of that object-transport layer.

Accordingly, the central question of this note is:

Given a declared replay-artifact graph, declared local objects, and induced directed transports, can one read inconsistency as explicit cycle non-closure rather than as a scalar plausibility deficit?

The rest of the note answers this question in a minimal equal-rank nonsingular setting, while keeping the public formulation narrow, reproducible, and bounded.

2 Observation Boundary

The present formulation is intentionally protocol-bounded. We do not attempt to expose or model an entire hidden-state trajectory of a language model. Instead, we work on a declared *replay-artifact graph*, whose nodes and relations are generated by a reproducible replay procedure.

2.1 Replay-artifact graph

Let

$$G = (V, E)$$

be a finite directed graph whose vertices are declared replay artifacts and whose edges are declared inter-artifact relations. Typical vertex types may include token-state artifacts, support artifacts, conflict artifacts, and answer-state artifacts, provided that their production is governed by a declared replay protocol.

The graph is not assumed to be a universal semantic graph. It is a protocol-defined observation boundary. Its purpose is to expose a finite relational substrate on which transport and closure can be read in a reproducible way.

Each directed edge

$$(v \rightarrow u) \in E$$

declares that the local object at artifact v is to be compared in the local coordinate system of artifact u . Direction therefore belongs to the protocol and is not inferred after the fact.

2.2 Declared cycle family

The basic observables in this note are not arbitrary graph statistics. They are read only on a declared family of explicit directed cycles

$$\mathcal{C} = (v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_{k-1} \rightarrow v_0).$$

A cycle is part of the observation boundary only if all of its directed legs are explicitly declared by the replay protocol. No implicit closure edge is invented for convenience.

When tree-chord or reference-skeleton constructions are used, the resulting cycle is admissible only if every leg of the induced cycle is itself declared by the protocol, or if an inverse-transport convention is explicitly declared. In the equal-rank nonsingular setting, such a convention uses $R_{v \leftarrow u} := R_{u \leftarrow v}^\top$ for the reversed leg. Thus, closure is always evaluated on an explicit protocol object rather than on an informal geometric sketch.

2.3 Reference skeleton and residual relations

Some evaluation procedures may also define a protocol-selected reference skeleton inside the replay-artifact graph, for example by selecting a high-agreement spanning structure. Such a skeleton is not assumed to be *true*. It is only a declared reference structure against which residual relations may be read.

This distinction is important. The note does not assume that a selected tree-like backbone represents ground truth. It assumes only that the protocol may designate a reference skeleton in a reproducible way, and that explicit residual relations relative to that skeleton may then be analyzed as potential closure-bearing or closure-breaking structure.

2.4 Why this boundary is narrow

This boundary excludes several stronger ambitions.

First, it excludes graph-wide smoothing or field-style aggregation as a public primitive. Second, it excludes any claim that every relevant inconsistency must appear on the declared graph. Third, it excludes undeclared post hoc intervention outside the replay protocol.

What it preserves is a narrower and cleaner claim: a replay-artifact graph can be constructed by a declared, reproducible procedure from declared local objects and declared inter-artifact relations, and explicit cycle-based transport behavior can be read on that graph without appealing to a universal semantic state space.

3 Local Objects

3.1 Projector-first public object

At each vertex $v \in V$, we associate a local linear object

$$\mathcal{O}_v := (P_v, \Sigma_v),$$

where P_v is an orthogonal projector onto a local subspace of $\mathbb{R}^d$, and Σ_v is an optional retained singular-value spectrum describing local anisotropy or concentration. The projector P_v is the primary public primitive. Any basis used to realize P_v is auxiliary.

Given a protocol-extracted local feature matrix

$$X_v \in \mathbb{R}^{d \times m_v},$$

let U_v denote the retained left singular frame under a declared rank rule, and let Σ_v denote any retained singular-value spectrum. In the equal-rank main-text setting,

$$U_v \in \mathbb{R}^{d \times r}$$

has orthonormal columns spanning the local subspace at vertex v , so that

$$U_v^\top U_v = I_r.$$

Then

$$P_v := U_v U_v^\top.$$

The basis U_v is not unique. If $G_v \in O(r)$, then

$$U'_v := U_v G_v \quad \Rightarrow \quad U'_v U'^\top_v = U_v U_v^\top = P_v.$$

Thus the projector component of the public local object is unchanged under a local change of gauge; any retained singular-value spectrum is likewise basis-independent.

This projector-first choice is the key observational discipline of the note. It separates what is publicly meaningful from what is merely a choice of local coordinates inside the same subspace.

3.2 Equal-rank main-text setting

For the main exposition, we assume an equal-rank nonsingular setting: all local objects have the same rank r , and pairwise overlap matrices used for transport are nonsingular. This assumption is not claimed as a universal property of implementation. It is adopted here because it isolates the geometric structure of transport and closure from the bookkeeping burden of rank variation.

Under this assumption, each vertex v carries an orthogonal frame

$$U_v \in \mathbb{R}^{d \times r}, \quad U_v^\top U_v = I_r,$$

with associated projector $P_v = U_v U_v^\top$.

3.3 Overlap matrices

For a directed edge $(v \rightarrow u)$, define the overlap matrix

$$C_{u \leftarrow v} := U_u^\top U_v \in \mathbb{R}^{r \times r}.$$

This matrix records how the local frame at v appears inside the local coordinate system of u .

The overlap itself is not yet the transport. Rather, it is the raw comparison object from which both edge-level misalignment and directed transport will be derived.

4 Directed Transport and Edge-Level Misalignment

4.1 Edge-level misalignment

We use the Frobenius-energy loss

$$\delta_{u \leftarrow v}^{\text{edge}} := r - \|C_{u \leftarrow v}\|_F^2$$

as an edge-level misalignment readout. Since

$$\|C_{u \leftarrow v}\|_F^2 = \sum_{i=1}^r \sigma_i(C_{u \leftarrow v})^2,$$

this quantity vanishes exactly when the two r -dimensional subspaces agree.

Although indexed by a directed relation, this edge loss is symmetric in the two subspaces. Direction enters not through the scalar loss itself, but through the ordered transport factors that compose around a cycle.

4.2 Directed transport via the polar factor

The directed transport from v into u is defined from the orthogonal polar factor of the overlap:

$$R_{u \leftarrow v} := \text{polar}(C_{u \leftarrow v}) \in O(r),$$

whenever $C_{u \leftarrow v}$ is nonsingular.

Equivalently, if

$$C_{u \leftarrow v} = A_{uv} \Sigma_{uv} B_{uv}^\top, \quad \Sigma_{uv} > 0,$$

is the singular value decomposition of the overlap, then

$$R_{u \leftarrow v} = A_{uv} B_{uv}^\top.$$

This is the closest orthogonal map to the overlap in Frobenius norm. It is therefore the natural directed transport factor attached to the declared edge in the equal-rank nonsingular setting; see Schönemann for the orthogonal Procrustes alignment perspective and Higham for the polar decomposition perspective [2, 3].

Two remarks are important.

First, $R_{u \leftarrow v}$ is not inferred from a scalar score. It is constructed directly from the pair of local subspaces.

Second, transport is genuinely directional. Even when edge-level misalignment is symmetric as a subspace comparison quantity, the ordered composition of transports around a cycle is not. This ordered composition is exactly where closure defect enters.

4.3 Reference-skeleton reading

If a protocol-selected reference skeleton is present, then the transport law is unchanged. What changes is only which edges are read as backbone relations and which are read as residual relations relative to that backbone. The tree is therefore not assumed to be true; it is a protocol-selected reference skeleton, often selected by a declared high-agreement rule under the protocol.

5 Explicit-Cycle Holonomy and Closure Defect

5.1 Cycle holonomy

Let

$$\mathcal{C} = (v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_{k-1} \rightarrow v_0)$$

be a declared directed cycle on the replay-artifact graph. For each directed leg $(v_i \rightarrow v_{i+1})$, with indices understood modulo k , the induced transport from the local coordinates of v_i into those of v_{i+1} is

$$R_{v_{i+1} \leftarrow v_i}.$$

The cycle holonomy is the ordered product

$$W_C := R_{v_0 \leftarrow v_{k-1}} R_{v_{k-1} \leftarrow v_{k-2}} \cdots R_{v_1 \leftarrow v_0} \in O(r).$$

This operator maps the local coordinates of v_0 back into themselves after transport around the full declared cycle.

In this formulation, perfect closure along the declared cycle corresponds to $W_C = I_r$. If the cycle fails to close, the departure of W_C from the identity records that failure.

5.2 Closure defect

We therefore define the closure defect of a declared cycle by

$$\Delta_C := \|W_C - I_r\|_F,$$

or, when a scale-comparable form is preferred,

$$\bar{\Delta}_C := \frac{1}{\sqrt{2r}} \|W_C - I_r\|_F.$$

The normalized form is introduced only as a comparable-scale readout; it is not claimed to be a universal normalization to the unit interval. In the equal-rank nonsingular setting, changing the root of the same directed cycle conjugates W_C by the corresponding transport product, and therefore leaves both Δ_C and $\bar{\Delta}_C$ unchanged.

These quantities are not edge-level losses. They are cycle-level non-closure observables. Their meaning is therefore distinct from the local overlap loss $\delta_{u \leftarrow v}^{\text{edge}}$. An edge may exhibit nontrivial misalignment while a larger cycle still closes, and conversely a cycle may fail to close even when no single edge appears extreme in isolation.

5.3 Why closure is different from scalar scoring

The point of Δ_C is not that it is another score. Its role is to read a different object.

A scalar plausibility score is attached to an item. An edge loss is attached to a pairwise subspace comparison. A closure defect is attached to an explicit declared cycle and therefore to an ordered relational structure. It answers a different question:

If one transports local structure around this declared cycle, does one return to where one started?

This is the minimal sense in which the present telemetry is transport-first. The primitive failure is not low plausibility. It is non-closure under declared transport.

5.4 Reference skeletons and residual chords

When a protocol-selected reference skeleton is present, a residual relation may form a declared cycle together with a declared path on the skeleton. In that case, the same holonomy definition applies, provided that every leg of the resulting cycle is declared by the protocol, or that an inverse-transport convention has been explicitly declared.

Thus tree-chord analysis does not introduce a second notion of defect. It simply provides one family of declared cycles on which the same closure observable can be evaluated.

5.5 Equal-rank scope

The holonomy definition above is stated only for the equal-rank nonsingular setting, where each transport factor is an orthogonal matrix in $O(r)$. Rank-aware transport is handled separately in Appendix A by partial isometries. The main text stays in the equal-rank case so that closure defect can be read without mixing geometric structure with rank bookkeeping.

6 Gauge Covariance and Gauge-Invariant Closure Residual

6.1 Local gauge action

For each node v , let a local orthogonal gauge change be given by

$$U'_v = U_v G_v, \quad G_v \in O(r).$$

By construction, the projector remains unchanged:

$$P'_v = U'_v U'^{\top}_v = P_v,$$

so the public projector-valued local object is gauge-invariant.

Under this basis change, the overlap matrix transforms as

$$C'_{u \leftarrow v} = U'^{\top}_u U'_v = G_u^{\top} C_{u \leftarrow v} G_v.$$

In the equal-rank nonsingular case, the orthogonal polar factor is equivariant under this transformation, so the induced transport obeys

$$R'_{u \leftarrow v} = G_u^{\top} R_{u \leftarrow v} G_v.$$

6.2 Cycle covariance

Let

$$\mathcal{C} = (v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_{k-1} \rightarrow v_0)$$

be a declared cycle, and let

$$W_{\mathcal{C}} = R_{v_0 \leftarrow v_{k-1}} R_{v_{k-1} \leftarrow v_{k-2}} \cdots R_{v_1 \leftarrow v_0}$$

be its holonomy.

Under local gauge transforms $G_{v_i} \in O(r)$, each factor transforms by left and right multiplication by the gauges at its target and source vertices, and the intermediate gauge terms cancel in the ordered product. Therefore

$$W'_{\mathcal{C}} = G_{v_0}^{\top} W_{\mathcal{C}} G_{v_0}.$$

So the cycle holonomy is gauge-covariant by conjugation at the root vertex of the cycle.

6.3 Gauge-invariant defect

Because the cycle holonomy transforms by conjugation at the root, any orthogonally conjugation-invariant distance from the identity gives a gauge-invariant closure readout. We record the Frobenius case as the following proposition.

Proposition 1 (Gauge-invariant closure defect). *In the equal-rank nonsingular setting, the closure defect of any declared cycle is invariant under independent local orthogonal basis changes at its vertices.*

Proof. Projectors are unchanged under $U_v \mapsto U_v G_v$. The overlap transforms as

$$C'_{u \leftarrow v} = G_u^{\top} C_{u \leftarrow v} G_v.$$

By equivariance of the orthogonal polar factor in the nonsingular equal-rank case,

$$R'_{u \leftarrow v} = G_u^{\top} R_{u \leftarrow v} G_v.$$

Multiplying the transformed transports around a declared cycle gives

$$W'_{\mathcal{C}} = G_{v_0}^{\top} W_{\mathcal{C}} G_{v_0},$$

since all intermediate gauge factors cancel. Frobenius norm is invariant under orthogonal conjugation, hence

$$\|W'_{\mathcal{C}} - I_r\|_F = \|W_{\mathcal{C}} - I_r\|_F.$$

The same holds for the scale-comparable normalized form $\bar{\Delta}_{\mathcal{C}}$. □

6.4 Interpretive consequence

The significance of Proposition 1 is practical as well as formal. It means that the closure residual does not depend on arbitrary local basis choices inside each retained subspace. What is being measured is therefore not a coordinate artifact but a cycle-structural failure of transport consistency on the declared replay graph.

This is the strongest invariance claim made in the note. We do not claim invariance under arbitrary graph rewrites, arbitrary rank changes, or graph-wide operator intervention. We claim only that, once the replay-artifact graph and local projector objects are fixed, the equal-rank cycle closure residual is invariant under local orthogonal gauge choice.

7 Bounded Empirical Status

The mathematical construction above is motivated by a bounded empirical program in which replay-artifact graphs, local projector objects, directed transports, and explicit cycle readouts are constructed by a declared protocol and evaluated under declared comparison conditions.

The empirical reading used in the present note is deliberately narrow:

Within a declared replay boundary, transport-first closure readouts can be computed reproducibly on explicit protocol-defined cycles, and these readouts can yield localized diagnostic structure that is not reducible to a scalar confidence replacement.

This note therefore does *not* present a universal detector. What it presents is a bounded empirical readout whose role is to report which declared cycle fails to close, under which declared comparison condition, and relative to which declared reference structure.

7.1 What is empirically earned

The bounded empirical claims stated below summarize a protocol-specific evidence base and are included here to delimit the interpretation of the formalism, not to serve as a stand-alone empirical study [1].

(i) Closure readout is operationally computable. For declared replay-artifact graphs and declared cycle families, the induced transport factors and resulting closure defects can be computed reproducibly by a declared procedure. This is the minimum operational claim required for the transport-first formulation to be meaningful.

(ii) The readout is localized rather than purely scalar. The resulting telemetry is not merely a replacement scalar for confidence. It identifies cycle-specific non-closure on declared graph structure. In this sense, the empirical readout is localized: it reports where closure fails, not only whether some global item appears weak.

(iii) The empirical signal is protocol-bounded. The observed behavior depends on the declared replay boundary, declared local objects, declared cycle family, and declared comparison conditions. The note therefore interprets empirical results only within bounded protocol settings rather than as evidence for unrestricted domain-general performance.

(iv) Negative and mixed outcomes are part of the record. The current empirical record includes not only positive readouts but also mixed and negative comparisons. This matters because the present formulation is intended as a bounded observation protocol rather than as a selectively positive summary. A transport-first readout that does not improve a comparison condition remains part of the empirical record.

7.2 How this differs from confidence replacement

It is important to state explicitly what is *not* being claimed.

The empirical role of Δ_C is not:

- to replace token confidence,
- to replace likelihood,
- to replace all scalar agreement measures, or
- to certify correctness by itself.

Instead, its empirical role is diagnostic and structural. A closure defect is useful precisely when the protocol requires one to ask not only whether an output appears plausible, but whether a declared relational cycle remains transport-consistent under the induced geometry of the replay graph.

7.3 Bounded persistence and comparison discipline

Some transport-first readouts may remain informative across more than one protocol-defined evaluation slice or comparison condition, while others may fail to do so. The present note does not promote persistence beyond what has been explicitly earned under declared comparison conditions.

Accordingly, the empirical interpretation remains bounded:

- informative under some protocol-defined evaluation slices,
- unresolved under others,
- and not entitled to regime-invariant language unless separately demonstrated.

This bounded reading is intentional. It keeps the observational primitive tied to the empirical record and prevents the formalism from claiming more stability than the evidence warrants.

8 Limits and Non-Claims

The present formulation is intentionally narrow. Its purpose is to define a transport-first observation protocol on a replay artifact graph, not to solve inconsistency detection in general. The limits stated here are part of the note itself.

8.1 Bounded protocol scope

All claims in this note are conditional on a declared replay boundary. This includes:

- a declared replay-artifact graph,
- declared local objects,
- declared directed relations,
- declared cycle families,
- and declared comparison conditions.

Nothing in the note implies that every meaningful inconsistency must appear inside such a boundary, or that a single protocol choice is sufficient for all settings.

8.2 No universal detector claim

This note does not claim a universal detector of inconsistency, hallucination, or error. It does not claim that every globally inconsistent output will necessarily induce a large closure defect on every declared cycle. It claims only that explicit-cycle closure residuals provide a bounded and gauge-invariant structural readout once the replay protocol has fixed what is to be observed.

8.3 No graph-wide operator or graph-wide smoothing claim

The note does not introduce graph-wide operators, graph-wide smoothing, or graph-wide aggregation as part of its public formulation. All observables are local-object and explicit-cycle based. The mathematical core therefore remains at the level of projector-valued local objects, induced edge transport, and closure residuals on declared cycles.

8.4 No mechanism-complete claim

The note defines a closure observable. It does not claim to have settled the full mechanism by which all empirical failure modes arise. In particular, the existence of a useful closure-defect readout should not be confused with a full theory of why every comparison pattern occurs. The formalism identifies a transport-level structural failure; it does not by itself complete every explanatory bridge.

8.5 No unrestricted portability claim

Even when a transport-first readout remains informative across more than one declared comparison condition, this note does not promote that observation into an unrestricted portability claim. Any persistence beyond a bounded protocol setting must be earned separately.

8.6 Equal-rank limitation of the main text

The main text works in the equal-rank nonsingular setting. This is a deliberate simplification made in order to state transport, holonomy, and gauge covariance cleanly. Rank-aware transport is relevant in practice, but it is treated separately in Appendix A through partial isometries rather than folded into the main development.

8.7 Auxiliary basis only

The bases U_v are construction devices. Public claims attach to the projectors P_v , optional basis-independent singular-value data, and gauge-invariant cycle residuals.

8.8 No implicit-cycle convenience

The closure formalism applies only to explicit declared cycles. No implicit closure edge is invented merely to make a geometric picture look complete. If a cycle is read through a reference skeleton and a residual relation, then every leg of the induced cycle must itself be declared by the protocol, or an inverse-transport convention must be explicitly declared.

8.9 Core claim

After all of these restrictions, the core claim can be stated plainly:

A replay-artifact graph equipped with projector-first local objects and induced directed transports admits an explicit-cycle closure residual that is gauge-invariant in the equal-rank nonsingular case and empirically readable within bounded protocol-defined comparison settings.

That claim is intentionally smaller than a universal theory. Its purpose is to provide a precise, reproducible, and mathematically legible observation layer on which later work may build.

9 Minimal Conclusion

The main contribution of this technical note is a disciplined reformulation of replay evaluation around four primitives:

- local objects,
- transport,
- explicit-cycle closure,
- gauge-invariant defect observables.

This is a minimal layer for discussing locally plausible but globally inconsistent behavior without reducing the observation problem to score-first heuristics.

Appendix A. Rank-Aware Transport

The equal-rank main text assumes that each transport factor lies in a single orthogonal group $O(r)$. In implementation, neighboring local objects may instead have unequal ranks or rank-deficient overlap. In that case we define

$$C_{u \leftarrow v} := U_u^\top U_v \in \mathbb{R}^{r_u \times r_v}$$

and take its compact SVD

$$C_{u \leftarrow v} = \tilde{U}_{uv} \tilde{\Sigma}_{uv} \tilde{V}_{uv}^\top.$$

The rank-aware transport primitive is the overlap-supported partial isometry

$$Q_{u \leftarrow v} := \tilde{U}_{uv} \tilde{V}_{uv}^\top \in \mathbb{R}^{r_u \times r_v}.$$

If the retained overlap rank is zero, the protocol records a null supported transport for that edge rather than an orthogonal factor or a nontrivial partial isometry. In the unequal-rank case, the transport object $Q_{u \leftarrow v}$ is a partial isometry on the overlap-supported subspace, not an orthogonal matrix and not an element of a single $O(r)$. Accordingly, rank-aware cycle composition is not written as a direct product of same-size orthogonal factors. Instead, after a declared root vertex is fixed, the transported maps are composed into a square endomorphism on the root coordinate space of the cycle. For a declared cycle rooted at v_0 , the rank-aware cycle map is

$$W_{\mathcal{C}}^{Q, v_0} := Q_{v_0 \leftarrow v_{k-1}} Q_{v_{k-1} \leftarrow v_{k-2}} \cdots Q_{v_1 \leftarrow v_0} \in \mathbb{R}^{r_{v_0} \times r_{v_0}}.$$

A corresponding rank-aware non-closure readout is

$$\Delta_{\mathcal{C}}^{Q, v_0} := \|W_{\mathcal{C}}^{Q, v_0} - I_{r_{v_0}}\|_F.$$

This readout may include both angular non-closure and rank-support loss. However, this appendix does not claim a gauge-invariance theorem of the same strength as Proposition 1. The unequal-rank construction is recorded here as a rank-aware implementation summary rather than as a main-text extension.

Thus the notation is intentionally split:

- $R_{u \leftarrow v}$: equal-rank nonsingular orthogonal transport from Section 4,
- $Q_{u \leftarrow v}$: rank-aware overlap-supported partial isometry.

The implemented protocol uses the rank-aware form when required, even though the main propositions in the note remain stated in the equal-rank nonsingular case.

References

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